SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I (Ajay J / 192511016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

SOLUTIONS

1. Doctor Shift Scheduling Using Backtracking Search
Problem

Assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) satisfying the following constraints:

- D1 cannot work Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work Morning shift.
- Every doctor must be assigned exactly one shift.

Step 1: Initial State
Doctors: D1, D2, D3
Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Initially, no doctor is assigned.

Step 2: Backtracking Search
Assign D1
Try Morning ✓
Assignment:

- D1 → Morning

No constraint violated.

Assign D2
Morning is already occupied.
Try Afternoon ✓
Assignment:

- D1 → Morning
- D2 → Afternoon

No constraint violated.

Assign D3
Morning (Not allowed)
Afternoon (Already occupied)
Night ✓
Check constraint:
D2 (Afternoon) comes before D3 (Night).
Constraint satisfied.

Final Assignment

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Check

Constraint	Status
D1 cannot work Night	Satisfied

Constraint	Status
D2 before D3	Satisfied
One doctor per shift	Satisfied
D3 cannot work Morning	Satisfied
Every doctor assigned once	Satisfied

Final Schedule

- Morning → D1
- Afternoon → D2
- Night → D3

2. Robot Navigation Using Breadth-First Search (BFS)

Grid (5 × 5)

```
S  .  .  X  .
.  X  .  X  .
.  X  .  .  .
.  .  X  X  .
.  .  .  .  G
```

Legend:

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Movement Allowed:

- Up
- Down
- Left
- Right

Cost per move = 1

Heuristic = Manhattan Distance (for reference)

BFS Node Expansion

Step 1

Queue:

[S]

Expand:

S

Visited:

S

Step 2

Add reachable neighbours.

Queue:

(0,1)

(1,0)

Step 3

Expand (0,1)

Queue:

(1,0)

(0,2)

Step 4

Expand (1,0)
Queue:
(0,2)
(2,0)

Step 5
Expand remaining nodes level by level.
Continue until Goal is reached.

Optimal Path
S
↓

↓

→

→

↓

↓

→

G
One possible path:
(0,0)
↓

(1,0)
↓

(2,0)
→

(2,1)
→

(2,2)
↓

(3,2)
↓

(4,2)
→

(4,3)
→

(4,4)=Goal

Path Cost
Each move = 1
Total moves = 8
Total Cost = 8

3. Autonomous Rescue Robot

(a) Constraint Analysis
1. Four-direction Movement
The robot can move only:

- Up
- Down
- Left
- Right

Diagonal movement is not allowed.

2. Obstacles
Blocked rooms cannot be entered.
The robot must find an alternative route.

3. Limited Sensing
The robot observes only adjacent cells.
It cannot know the entire map initially.

4. Movement Cost
Each movement costs 1.
The robot tries to minimize total travel cost.

5. Risky Zones
Entering risky areas adds 2 extra cost.
The robot prefers safer paths whenever possible.

6. Dynamic Environment
As the robot explores, it updates its map.
Future decisions are based on newly discovered information.

(b) Solution Approach
Search Strategy
Uniform Cost Search (UCS) with Online Search Agent
Steps

1. Start at S.
2. Observe neighboring cells.
3. Store discovered information.
4. Calculate total path cost.
5. Expand the lowest-cost node.
6. Avoid blocked paths.
7. Avoid risky areas when possible.
8. Continue until Goal (G) is reached.

(c) AI Concepts
Intelligent Agent

The rescue robot senses its surroundings and chooses the best action automatically.

Rationality
The robot always selects the action with the lowest total cost while safely reaching the survivor.

Environment Type
The environment is:
- Partially Observable
- Dynamic
- Sequential
- Discrete
- Single-Agent

(d) Justification
Uniform Cost Search is the most suitable algorithm because:
- It always finds the minimum-cost path.
- It considers movement cost and risky zones.
- It works well when path costs are different.
- Combined with an Online Search Agent, it can update decisions whenever new information is discovered.
- It enables safe and efficient navigation in an unknown environment.

Conclusion
- Question 1: Backtracking Search successfully assigns doctors while satisfying all constraints.
- Question 2: BFS explores nodes level by level and finds the shortest path with a total cost of 8.
- Question 3: Uniform Cost Search combined with an Online Search Agent provides the safest and least-cost path for the rescue robot in a dynamic, partially observable environment.